Project Defense Preparation

AI-Based Cyclone Early Warning System

Master's Thesis · Bay of Bengal Intensification Prediction · 2026

This document contains every question a professor may ask during the defense, with nested follow-up questions and complete answers. Organized by topic.

CATEGORY 1: PROJECT OVERVIEW & MOTIVATION

Q1. What is your project about?

Answer: This project is an AI-based early warning system that predicts whether an existing tropical cyclone in the Bay of Bengal will significantly intensify — specifically, whether wind speed will increase by 15 knots or more — within two forecast windows: 48–72 hours (2–3 days) and 120–168 hours (5–7 days).

■ ***Follow-up: Why focus on intensification rather than cyclone formation?***

Formation prediction requires different models (genesis indices). Intensification is operationally more critical — a storm can form as a weak depression and rapidly strengthen before landfall, giving less time to warn coastal populations. Operational agencies like the India Meteorological Department specifically focus on intensification forecasts.

■ ***Follow-up: Why Bay of Bengal specifically?***

The Bay of Bengal has disproportionately high cyclone mortality compared to other ocean basins. The funnel-shaped coastline of Bangladesh, shallow bathymetry increasing storm surge, and dense coastal population make accurate early warning especially critical. Cyclones like Sidr (2007), Fani (2019), and Amphan (2020) caused thousands of deaths and billions in damage.

Q2. What is the threshold of 15 knots — why that number?

Answer: 15 knots (~28 km/h) was chosen as a research-defined intensification threshold. The standard Rapid Intensification (RI) definition used by the National Hurricane Center is 30 knots in 24 hours. We used 15 knots over a longer window (48–72 hours) to make the label more applicable to the Bay of Bengal where storms intensify at different rates than Atlantic hurricanes.

■ ***Follow-up: Is this threshold scientifically validated?***

It is a research parameter, not an operational standard. For a thesis, it is clearly stated as a project-defined threshold. Future work could test sensitivity to different thresholds (10 kt, 20 kt, 30 kt) to validate the choice.

■ ***Follow-up: What happens if you change the threshold?***

A lower threshold (e.g., 10 kt) would create more positive labels — the model would be trained on more 'intensifying' events. A higher threshold (e.g., 30 kt) would create fewer positives, increasing class imbalance and making the problem harder. The 15 kt threshold gives reasonable class balance for the Bay of Bengal dataset.

CATEGORY 2: DATA & DATASETS

Q1. What data did you use?

Answer: Five data sources: (1) IBTrACS v04r01 from NOAA — 375 Bay of Bengal storms from 1970–2024, 6-hourly track observations. (2) ERA5 reanalysis from Copernicus CDS — atmospheric variables (SST, pressure, wind, temperature) for 2018–2020. (3) GridSat-B1 satellite infrared images from NOAA AWS S3 — 445 images from 9 major cyclones. (4) Open-Meteo API — live weather for the real-time dashboard. (5) GDACS RSS feed — active

cyclone alerts.

■ ***Follow-up: What is IBTrACS?***

IBTrACS stands for International Best Track Archive for Climate Stewardship. It is maintained by NOAA and contains best-track estimates for all global tropical cyclones since 1842, compiled from multiple meteorological agencies including the India Meteorological Department, Joint Typhoon Warning Center, and WMO.

■ ***Follow-up: What is ERA5?***

ERA5 is the fifth generation atmospheric reanalysis from the European Centre for Medium-Range Weather Forecasts (ECMWF), produced by the Copernicus Climate Change Service. It reconstructs historical atmospheric conditions at 0.25 degree spatial resolution by combining model output with observations using data assimilation. It covers 1940 to present.

■ ***Follow-up: What is a reanalysis — is it real observation data?***

No, reanalysis is a model-based reconstruction. It uses numerical weather prediction models combined with historical observations to produce a physically consistent estimate of past atmospheric conditions. It is more complete than raw observations but contains model errors. For operational forecasting, it would be replaced with NWP forecast fields.

■ ***Follow-up: Why only 2018–2020 for ERA5?***

ERA5 files are large (~163 MB per year) and require a Copernicus account. We downloaded 3 years as a proof of concept. The V2 model was trained on 21 storms from those 3 years. Extending to 2010–2020 is identified as future work.

Q2. What are the GridSat-B1 satellite images?

Answer: GridSat-B1 provides 3-hourly global infrared brightness temperature images from geostationary satellites, available from NOAA's AWS S3 bucket at no cost. The infrared window channel at 11 micrometers captures cloud-top temperatures. Cold (bright) pixels indicate high cloud tops which correspond to deep convective activity — the physical signature of an intensifying cyclone.

■ ***Follow-up: Why infrared and not visible light images?***

Infrared images work 24 hours a day — they do not depend on sunlight. Visible images are only available during daylight. For a cyclone early warning system, 24/7 availability is essential.

■ ***Follow-up: Why 128×128 or 64×64 pixels?***

The patch size was chosen to capture the full cyclone structure (approximately ±8 degrees around the storm center) while remaining computationally manageable. At 0.07 degree GridSat resolution, 8 degrees = ~115 pixels, which maps to 128 after interpolation.

■ ***Follow-up: What does the CNN see in these images?***

The CNN learns spatial patterns associated with intensification: symmetric circular banding of cold cloud tops, formation of a clear eye, cold ring structure (eye wall), and overall cloud-top temperature. These features are physically meaningful — they correspond to the organized convection that drives intensification.

CATEGORY 3: MACHINE LEARNING METHODOLOGY

Q1. What machine learning models did you use?

Answer: Three versions. V1 and V2 use XGBoost — an ensemble of gradient-boosted decision trees. V3 uses a CNN-LSTM neural network implemented in PyTorch. XGBoost was chosen for V1/V2 because it handles tabular data well, is robust to missing values, and produces interpretable feature importances. CNN-LSTM was chosen for V3 to process spatiotemporal image sequences.

■ *Follow-up: What is XGBoost?*

XGBoost (Extreme Gradient Boosting) builds an ensemble of decision trees sequentially, where each new tree corrects the residual errors of the previous ones. It uses gradient descent in function space to minimize a loss function. Key advantages: handles missing values natively, fast, provides feature importance scores, and has strong regularization to prevent overfitting.

■ *Follow-up: What is a CNN?*

A Convolutional Neural Network (CNN) applies learnable filters across an image to detect spatial patterns. Each filter slides over the image and produces a feature map showing where that pattern is detected. Multiple layers detect increasingly abstract features — edges → shapes → circular cloud structures.

■ *Follow-up: What is an LSTM?*

A Long Short-Term Memory network is a type of recurrent neural network designed to learn temporal sequences. It uses gating mechanisms (input gate, forget gate, output gate) to selectively remember or forget information from previous time steps. In V3, it reads the sequence of CNN-extracted features from 4 consecutive satellite images to detect temporal evolution of the cyclone structure.

■ *Follow-up: Why combine CNN and LSTM?*

CNN alone processes each image independently and ignores temporal order. LSTM alone cannot process image data. Their combination allows the model to both extract spatial features from each image (CNN) and model the temporal evolution of those features over 18 hours (LSTM) — capturing whether the cloud structure is organizing or disorganizing.

Q2. How did you prevent overfitting?

Answer: Several techniques were used. For XGBoost: regularization parameters (max_depth=4, learning_rate=0.05), and storm-grouped train-test split prevents same-storm leakage. For CNN-LSTM: Dropout (0.3) after dense layers, Batch Normalization in CNN layers, early stopping (patience=15 epochs), weight decay (L2 regularization 1e-4), and class-weighted loss to prevent the model from predicting all positives.

■ *Follow-up: What is dropout?*

Dropout randomly sets a fraction of neuron activations to zero during training. This prevents neurons from co-adapting and forces the network to learn redundant representations. At test time, all neurons are active and outputs are scaled. With dropout=0.3, 30% of neurons are randomly disabled each training step.

■ *Follow-up: What is the GroupShuffleSplit and why is it important?*

GroupShuffleSplit is a scikit-learn cross-validator that ensures all observations from the same group (storm ID in our case) go to either train OR test — never split across both. Without this, the model could learn storm-specific patterns (e.g., Amphan's track) and appear to generalize when it is actually memorizing. This is the most critical methodological

safeguard in the project.

Q3. What is ROC-AUC and why did you use it?

Answer: ROC-AUC (Receiver Operating Characteristic Area Under Curve) measures the model's ability to rank positive (intensifying) examples above negative ones at all classification thresholds. A value of 1.0 means perfect ranking; 0.5 means random. We used it because it is threshold-independent — it evaluates the probability scores rather than binary predictions, which is appropriate for an early warning system where the operator may want to choose their own alert threshold.

■ *Follow-up: What is PR-AUC and when is it better than ROC-AUC?*

PR-AUC (Precision-Recall Area Under Curve) focuses on the positive class performance, making it more informative under class imbalance. If 80% of observations are non-intensifying, a model predicting all negatives gets high ROC-AUC but zero PR-AUC. Our dataset has ~74% positive class (intensifying) so ROC-AUC is reasonable, but we reported both.

■ *Follow-up: What is the Brier Score?*

The Brier Score measures the mean squared error between predicted probabilities and actual outcomes. A perfect model scores 0.0; random scoring 0.25 for balanced classes. Our V3 Brier Score of 0.143 indicates well-calibrated probability estimates — when the model says 70%, actual intensification happens approximately 70% of the time.

CATEGORY 4: RESULTS & PERFORMANCE

Q1. What were your results?

Answer: V1 (track only): ROC-AUC 0.5848 at 48-72h, 0.1000 at 120-168h. V2 (track + ERA5): 0.6696 at 48-72h, 0.8037 at 120-168h. V3 (track + satellite): 0.8862 at 48-72h. Both hypotheses were confirmed — each additional data modality improved prediction accuracy.

■ *Follow-up: Why is V1 so bad at 120–168h (0.10)?*

Track features alone carry almost no predictive information about what will happen 5–7 days ahead. The storm's current position and movement are determined by synoptic-scale steering currents that change completely over 5 days. Only large-scale atmospheric conditions (SST, wind shear, moisture) — which ERA5 provides — can predict long-range intensification potential.

■ *Follow-up: Why does V2 jump dramatically at 120-168h?*

SST and atmospheric instability are the dominant thermodynamic drivers of intensification. These conditions evolve slowly (days to weeks) compared to storm position, so a warm SST patch ahead of the storm's projected path is highly predictive of intensification 5–7 days out. This is consistent with Emanuel's (1986) maximum potential intensity theory.

■ *Follow-up: Why is V3 better than V2 at 48-72h?*

At 2–3 days, the cyclone's internal structure visible in satellite images becomes the most important predictor. A storm showing organized convection, developing eye wall, and symmetric banding is likely to intensify regardless of environmental conditions. The CNN-LSTM captures these structural signals that neither track statistics nor gridded reanalysis can directly observe.

■ ***Follow-up: Why couldn't you evaluate V3 at 120-168h?***

The 9-storm dataset used for V3 produced a test set (from GroupShuffleSplit) that contained only intensifying storms — all TARGET=1. With only one class in the test set, ROC-AUC cannot be computed (requires both classes). This is a dataset size limitation, not a model failure.

Q2. Your V2 training set only has 21 storms — isn't that too small?

Answer: Yes, this is the main limitation of V2. 21 storms from 2018–2020 is a small training set for a machine learning model. However, the LeaveOneGroupOut and GroupShuffleSplit evaluation still provides valid estimates. The results are preliminary — extending ERA5 coverage to 2010–2020 (approximately 100 storms) would be needed for production-quality V2 results.

■ ***Follow-up: How do you know the model isn't just overfitting to 21 storms?***

We used storm-grouped splitting so test storms are completely unseen during training. The V2 ROC-AUC of 0.80 at 120-168h on held-out storms (not seen during training) indicates genuine generalization. However, with 21 storms, the confidence interval is wide — results should be interpreted cautiously.

CATEGORY 5: LIVE DASHBOARD & SYSTEM

Q1. How does the live dashboard work?

Answer: The Streamlit dashboard has three integrated components. First, it checks GDACS for active Bay of Bengal cyclones — if found, it auto-fills the storm's position. Second, the user (or the system automatically) calls Open-Meteo API to fetch current SST, pressure, temperature, and wind for that position. Third, the trained V2 XGBoost model uses these live weather values plus the storm parameters to output 48-72h and 120-168h intensification risk probabilities.

■ ***Follow-up: What is the difference between V1 and V2 mode in the dashboard?***

V1 mode uses only the 9 sidebar sliders (lat, lon, wind, month, hour, lat change, lon change, wind change, track speed). V2 mode automatically activates when the user clicks 'Fetch Live Weather' — it adds SST, pressure, u/v wind, temperature, dewpoint, and precipitation from Open-Meteo to the prediction. The dashboard shows a green badge 'V2 (track + live weather)' when V2 is active.

■ ***Follow-up: Is Open-Meteo the same as ERA5?***

Not exactly. Open-Meteo provides current conditions from multiple NWP models at ~0.1° resolution. ERA5 is a historical reanalysis at 0.25°. They are not identical but cover the same physical variables. The V2 model was trained on ERA5 values, so using Open-Meteo introduces a small proxy bias. For a research prototype this is acceptable; an operational system would use ECMWF or GFS forecast fields.

■ ***Follow-up: What if there is no active cyclone — can the system still be used?***

Yes. The user can manually set any position using the lat/lon sliders to simulate a hypothetical scenario. For example, setting conditions matching Amphan 2020 at its peak (18.5°N, 87°E, 110 knots) would produce a high-risk prediction. This is also how the system was tested during development.

CATEGORY 6: LIMITATIONS & FUTURE WORK

Q1. What are the main limitations?

Answer: Five key limitations: (1) Intensification prediction only — no track, landfall, or flood prediction. (2) ERA5 is reanalysis, not real-time forecast — for true operational use, NWP forecast fields are needed. (3) Small ERA5 training set (21 storms). (4) V3 long-range evaluation was not possible due to dataset size. (5) Open-Meteo as ERA5 proxy introduces minor bias.

Follow-up: Why can't you predict flood from this?

Flood prediction requires a hydrological model combining storm track, rainfall rate, terrain elevation, river discharge, soil saturation, and coastal storm surge. Our model only outputs 'will the wind increase' — it says nothing about where the storm will go or how much rainfall will fall over land. These are separate, complex models.

Follow-up: What would you do differently if you had more time?

Three priorities: First, download ERA5 for 2010–2020 to give V2 ~100 training storms. Second, integrate INSAT-3D from MOSDAC for higher-resolution Indian Ocean satellite data. Third, replace the Open-Meteo proxy with operational GFS/ECMWF forecast fields to make V2 truly predictive rather than using current conditions.

Q2. How would you deploy this operationally?

Answer: An operational deployment would need: (1) Real-time NWP forecast fields replacing ERA5 and Open-Meteo. (2) Automated storm detection feeding directly from IMD or JTWC track data. (3) A production server running Streamlit or a REST API. (4) Alert dissemination system (SMS, email) connected to the risk output. The current system is a research prototype demonstrating the methodology.

Follow-up: Would this replace meteorologists?

No. The model is a decision support tool. Meteorologists at IMD or BMMD (Bangladesh Met Department) would review the AI risk output alongside their own analysis. The model provides probabilistic guidance — the final warning decision remains with trained experts. This is standard in operational AI-assisted forecasting.

CATEGORY 7: TECHNICAL DEEP DIVES

Q1. What is Haversine formula and why did you use it?

Answer: The Haversine formula calculates the great-circle distance between two points on a sphere given their latitudes and longitudes. We used it to compute TRACK_SPEED (storm translation speed in km/h) between consecutive 6-hourly IBTrACS observations. Simple Euclidean distance would be inaccurate at the Earth's curvature, especially at larger separations.

Q2. What is GroupShuffleSplit vs regular train-test split?

Answer: Regular train_test_split splits individual rows randomly. Since each storm has ~50 observations, a regular split would put some observations from AMPHAN in train and others in test — the model memorizes AMPHAN's track and appears to predict well. GroupShuffleSplit

assigns entire storms to either train or test, ensuring no storm is in both. This is essential for unbiased evaluation.

Q3. What is class imbalance and how did you handle it?

Answer: In our short-horizon dataset, ~74% of observations are labeled as intensifying (TARGET=1). A naive model predicting all 1s gets 74% accuracy but zero recall for the non-intensifying class. We handled this with: class-weighted loss in CNN-LSTM (BCEWithLogitsLoss with pos_weight), and PR-AUC as a supplementary metric that penalizes false positives against the positive class.

■ ***Follow-up: Did you try SMOTE or other resampling?***

No. SMOTE (Synthetic Minority Oversampling) generates synthetic samples for the minority class. We chose class weighting instead because SMOTE can create unrealistic synthetic cyclone states — a synthetic observation at lat=15 with wind=70 may be physically inconsistent. Class weighting adjusts the loss function without modifying the data distribution.

Q4. Why XGBoost over Random Forest?

Answer: Both were evaluated in V1. XGBoost consistently outperformed Random Forest (RF ROC-AUC ~0.65 vs XGBoost 0.67 for V2 short). XGBoost uses boosting (sequential error correction) vs RF's bagging (parallel independent trees). Boosting is generally superior for tabular data because later trees specifically target the hard examples the earlier trees got wrong.

CATEGORY 8: PRESENTATION & DEFENSE

Q1. How would you summarize your contribution in one sentence?

Answer: We built and validated a three-version AI system showing that adding ERA5 atmospheric reanalysis improves Bay of Bengal cyclone intensification prediction from ROC-AUC 0.58 to 0.80 at 5–7 days, and adding satellite infrared image sequences further improves it to 0.89 at 2–3 days.

Q2. What makes this project novel?

Answer: Three aspects: (1) Systematic ablation study comparing three data modalities on the same Bay of Bengal dataset — few published works directly compare track-only vs reanalysis vs satellite for BoB. (2) Live operational dashboard with automatic GDACS + Open-Meteo integration — most research stops at offline evaluation. (3) CNN-LSTM applied to GridSat-B1 historical archive for BoB — GridSat is underutilized compared to GOES/Himawari in existing deep learning work.

Q3. What would you tell your supervisor in 2 minutes?

Answer: We built an AI early warning system for Bay of Bengal cyclone intensification. Version 1 uses only historical track data — ROC-AUC 0.58. Version 2 adds ERA5 atmospheric reanalysis — jumps to 0.80 at 5-7 day prediction, proving thermodynamic conditions dominate long-range forecasting. Version 3 adds satellite infrared image sequences processed by a CNN-LSTM neural network — reaches 0.89 at 2-3 days, proving the cyclone's visible structure predicts short-range intensification better than any tabular feature. A live dashboard fetches

real storm positions from GDACS and real weather from Open-Meteo to produce zero-manual-input risk assessments. The code and all data are on GitHub.

— End of Defense Preparation Document —